Green Ghar Score: M1 Approach

Team Spades · Megathon 2026 · An explainable home energy-efficiency score and retrofit advisor for Indian homes and lenders

1 Problem statement and our understanding

Problem. The task is to give every home a transparent **Green Home Score (0–100)** from the data a lender or homeowner actually has. The score must show **why** the home scored as it did and **which retrofits** would raise it, with costs, savings and CO_2 impact. A conversational advisor explains the result in plain language, and a portfolio view lets a bank find, rank and finance its least efficient homes.

Our understanding: what makes this hard in India.

- **The energy problem is cooling, not heating.** Load is driven by cooling degree-days, AC star rating (ISEER), west-facing glass, roof heat gain and the AC set-point. Most Western scoring logic assumes heating-dominated homes and transfers badly.
- **Data is sparse and messy.** Most homes have no energy audit. What exists is a paper or PDF DISCOM bill (often in Hindi), partial appliance details, an optional BEE star label, and gaps in monthly readings. Self-reported data can contradict the meter.
- **Savings depend on the tariff.** Slab tariffs (telescopic, free-unit schemes, category switches) and net metering decide what a kWh saved is worth. LPG cooking and diesel backup carry their own emission factors.
- **Retrofits interact.** Fixing the envelope first shrinks what a new AC can save, so package savings are not the sum of standalone savings. Some "green" moves backfire: on today's coal-heavy grid (CEA 0.727 kg CO_2/kWh), switching LPG to induction *raises* CO_2 unless paired with solar.
- **Trust and compliance.** A lender cannot act on a black-box number or on an LLM that invents savings figures. Every number shown must trace back to a deterministic computation, and personal data must stay protected (DPDP Act 2023).

2 Idea / abstract

Green Ghar is a hybrid **physics + machine-learning** engine behind a **grounded multilingual advisor**.

- A deterministic India-specific energy model (climate zone, ISEER, envelope, appliances, cooking, backup, rooftop PV) estimates annual kWh and CO_2. A LightGBM model learns the residual between that estimate and real bills.
- The score is a **fixed, published, monotonic** weighted sum of seven sub-scores: energy efficiency 25%, consumption intensity 20%, envelope 15%, cooling system 15%, windows and shading 10%, renewables 10%, smart use 5%.
- Bands: 80+ Highly efficient · 65–79 Good · 50–64 Moderate · 30–49 Poor · <30 Very poor. Every score carries a confidence level and data-conflict warnings.
- Retrofits are simulated **sequentially** in interaction order, priced at the home's real slab tariff, ranked by a LambdaRank model, and turned into a payback roadmap with subsidies (PM Surya Ghar) and green-loan EMIs.
- A hedonic model with conformal intervals estimates the property-value uplift.
- An LLM advisor answers in **English, Hindi and Hinglish**, by text or voice, on Telegram. It only calls engine tools; a **numeric verifier** rejects any reply whose numbers do not match the tool outputs, and a template answer is used instead.
- For lenders, a portfolio API reports financed emissions (PCAF-style attribution), the worst homes, green-loan target lists and housing-society (RWA) common-solar leads.

Differentiators: bill photo → score (OCR); heatwave-readiness check; sequential, tariff-aware packages; a verified-number LLM; all of it built on free tiers.

3 High-level approach, pipeline and architecture

1 · Ingest — Form / API, bill photo (Gemini Flash → Tesseract OCR), voice (Whisper STT), Telegram

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2 · Validate — Pydantic schema, unit and range checks, missing-month imputation, label vs. meter conflict rule

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3 · Energy model — Physics (CDD base 24 °C, ISEER, envelope) × LightGBM residual → kWh, ₹, CO_2

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4 · Score — 7 monotonic sub-scores · fixed weights · band · confidence

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5 · Retrofits — Sequential simulation · slab tariff pricing · LambdaRank · roadmap and finance

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6 · Explain — LLM tool-calls → numeric verifier → EN/HI reply (text or TTS)

Channels	REST API (FastAPI, 33 routes) · Telegram bot (webhook) · voice in (Groq / NVIDIA Whisper) and out (NVIDIA Magpie TTS, edge-tts fallback) · bill-photo upload
Intelligence	Intent router (Hinglish, negation) → engine tools → LLM: NVIDIA NIM (primary) → Groq (fallback) → deterministic templates. The verifier handles lakh/crore, Indian digit grouping, Devanagari digits and % (0.5% tolerance).
Core engine	physics · consumption · score · tariff (state slabs, free units, net metering) · retrofits · solar · plan · finance · roadmap · heatwave · portfolio. Pure Python, deterministic, versioned reference data.
ML models	Energy: hybrid physics × LightGBM (log residual). Ranker: LightGBM LambdaRank. Value: hedonic LightGBM, monotone in score, split-conformal 80% intervals and 10 bootstrap models. Leakage guard on forbidden features.

Data	Supabase Postgres (ap-south-1): schemas *public* / *gg* (SQL port of the score and package engine, at parity with Python) / *synth* (in-database generator). Row-level security on every table, Auth JWT, a Storage bucket for bills, portfolio RPCs. Synthetic population of 50k+ homes calibrated to public sources (CEA, BEE, DISCOM tariffs, IMD climate, Census, NFHS).
Ops	Docker / compose, request-ID and timing middleware, uniform error envelope, OpenAPI spec, model and data cards, responsible-AI note, one-command rebuild (*make data train test*).

Key design decisions

- Physics first, ML second: explainable, monotonic and works with zero bills; ML corrects the bias where bills exist.
- Fixed weights instead of learned weights, so a lender can audit and reproduce every score.
- The LLM never computes. It only phrases tool outputs, and the verifier enforces this.

Data flow for one home

- Bill photo → 12 months of kWh and tariff → Home record → score with sub-scores and warnings.
- Top-k retrofit package → ₹/yr saved, CO_2 change, payback, score after → advisor reply.
- Score is stored in Supabase → appears in the lender's portfolio summary and ranking.

4 Planned testing methodology

What we test	Method	Pass criterion
Scoring correctness	Unit tests: the weights sum to 1, band edges, tariff slabs for each state; a hidden-scenario suite of hand-built archetype homes (e.g. old 1-star AC on a west-facing top floor vs. a 5-star inverter home with PV).	Exact expected bands; archetypes ordered correctly
Monotonicity / properties	Property tests: improving any single input (star rating, LED share, insulation, PV, set-point) must never lower the score or raise the kWh estimate.	0 violations across randomized homes
Energy and ranking models	Home-grouped 70/15/15 split (seed 42), so no home appears in both train and test. Compared against a median-by-zone/type baseline, physics-only and direct LightGBM. Ranker scored with NDCG@5 and Precision@3 against popularity and rule-ROI baselines. Leakage test on forbidden features.	Beat all baselines on the held-out test set; infeasible recommendations = 0%
Uncertainty	Empirical coverage of the split-conformal value intervals on test homes; confidence level reported for each score.	Coverage ≈ 80% ($\alpha = 0.2$)
Retrofit realism	Package savings vs. the sum of standalone savings (interaction check); the LPG → induction CO_2 sign check; payback against tariff hand calculations.	Package ≤ sum; signs physically correct
LLM groundedness	180 held-out prompts in 6 categories (explain score, explain a recommendation, what-if, infeasible request, missing data, conflicting data). Adversarial and Hinglish prompts, plus injected wrong numbers to confirm the verifier rejects them.	Groundedness 100%, unsupported claims 0%
Fairness	Score and error breakdown by city, climate zone, dwelling type, tenure and income band; error and score gaps reviewed before release.	No group MAPE gap > 5 pp
System and integration	API contract tests (all routes, error envelope), SQL ↔ Python parity fixtures for the score and package engines, bill-OCR fixtures, auth/RLS access tests, latency benchmarks.	Parity exact; p95 score < 300 ms, what-if < 500 ms

Status: pipeline implemented, 58 automated tests passing. Early held-out results: energy MAPE 9.7% vs. 26.0% for the baseline (R^2 0.94); ranker NDCG@5 0.986; template-mode groundedness 100%. Final numbers will be reported in later milestones.